



CORE

Center for Research in Energy
ECONOMICS AND MARKETS

Case Maker

Econometric Game 2026

ECONOMETRIC GAME 2026

CASE THEME

Energy trading and
forecasting of day-ahead
spot electricity prices



DEPARTMENT OF ECONOMICS
AND BUSINESS ECONOMICS
AARHUS UNIVERSITY



Center for Research in Energy
ECONOMICS AND MARKETS

ECONOMETRIC GAME 2026

NIELS HALDRUP
PROFESSOR



THE 2026 EG CASE MAKER TEAM



DEPARTMENT OF ECONOMICS
AND BUSINESS ECONOMICS
AARHUS UNIVERSITY



Center for Research in Energy
ECONOMICS AND MARKETS



Professor
Niels Haldrup
Jury member



Professor
Bent J. Christensen
Jury member



postdoc
Peter K. Christensen



postdoc
Thomas K. Kloster



postdoc
Gabriele Mingoli



postdoc
Anders M. Norlyk



DEPARTMENT OF ECONOMICS
AND BUSINESS ECONOMICS
AARHUS UNIVERSITY



Center for Research in Energy
ECONOMICS AND MARKETS

ECONOMETRIC GAME 2026

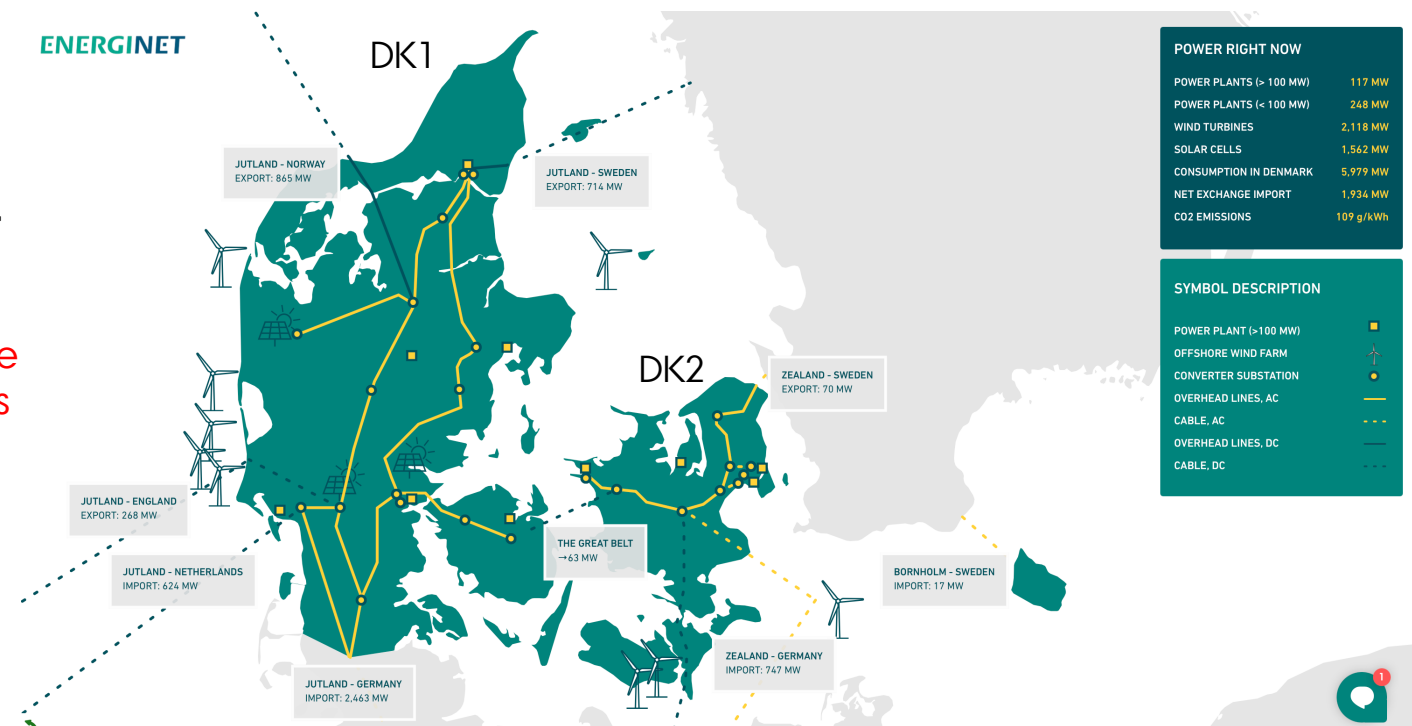
NIELS HALDRUP
PROFESSOR



CASE A: FORECASTING THE HOURLY DAY-AHEAD ELECTRICITY PRICES IN THE DK1 PRICE ZONE FOR 2024 - USING ONLY THE INFORMATION AVAILABLE AS OF 31 DECEMBER 2023

Forecasting tasks.

1. Forecasts of spot day-ahead prices for **each hour for each day of the first week of 2024** => 7x24 forecasts
2. Forecasts of the **weekly average of hourly spot day-ahead prices for every week of 2024** => 52x24 forecasts
3. Forecasting evaluations



<https://en.energinet.dk>

ABOUT THE SPOT DAY-AHEAD MARKET

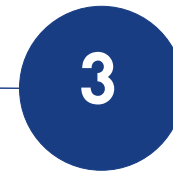
- The (spot) day-ahead electricity market at Nordpool is characterized by an auction at 12:00 CET each day, where all buy and sell orders for delivery each hour of the next day are collected and the equilibrium prices are determined
- Note that the hourly day-ahead prices are determined simultaneously, so the daily auction prices consist of a 24-dimensional vector for each day
- There is one market-clearing price per bidding zone and hour; all price zones are cleared simultaneously
- The market is characterized by a transmission infrastructure with capacity constraints that couples different price zones



10:00
Interconnector and grid
capacity published



12:00
Final buy and sell orders
submitted to the auction



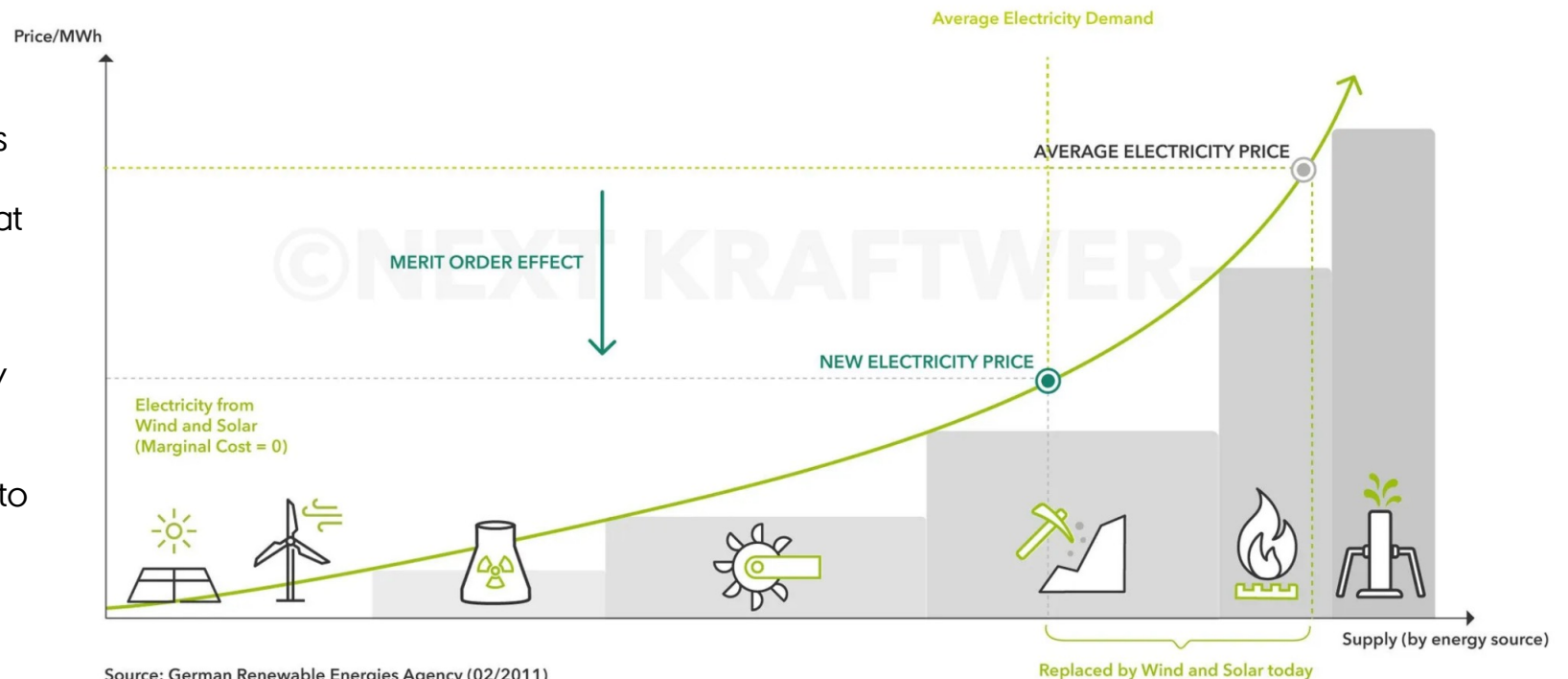
12:45
Market clearing prices
determined for each hour
and bidding zone



Next day, 00-24.
Prices become reference
for physical delivery and
further trading

ELECTRICITY SUPPLY IN MERIT ORDER AUCTIONS

- Electricity supply is determined by aggregation over marginal costs of energy sources
- Production by renewables (wind and solar with low marginal costs) has a great impact on equilibrium prices
- Note: demand is relatively inelastic, but still periodic
- Electricity is only storable to a limited degree => Price volatility



WHAT DETERMINES ELECTRICITY PRICES AND WHAT IS RELEVANT FOR FORECASTING OVER DIFFERENT TIME HORIZONS?

Data available for 2018-2023 to build forecasts for 2024.



FORECASTING TASKS



- **First week of 2024**
 - 7x24 hourly spot day-ahead forecasts
 - Only data up to and including 31/12/2023 may be used
 - High resolution, short horizon,
 - Strong operational relevance
- **Full year of 2024**
 - 52x24 weekly averages for each hour
 - Same information set
 - Lower resolution, longer horizon
 - Important for more strategic products, e.g. futures and forward contracts
- Both forecast sets must be evaluated against the observed 2024 prices – and part of the case is to discuss which forecasts are most relevant.
- **Note:**
 - Not all available data is of the same frequency
 - You are not expected to use all data – be selective and argue your choices
 - The point is not to choose a single model, but to decide what information is relevant for a particular forecasting horizon

ABOUT GAI, CLOUD COMPUTING ETC

Allowed use

- Help with coding tasks and debugging
- Optimization, and alternative coding
- Understanding programming concepts
- Generating coding snippets

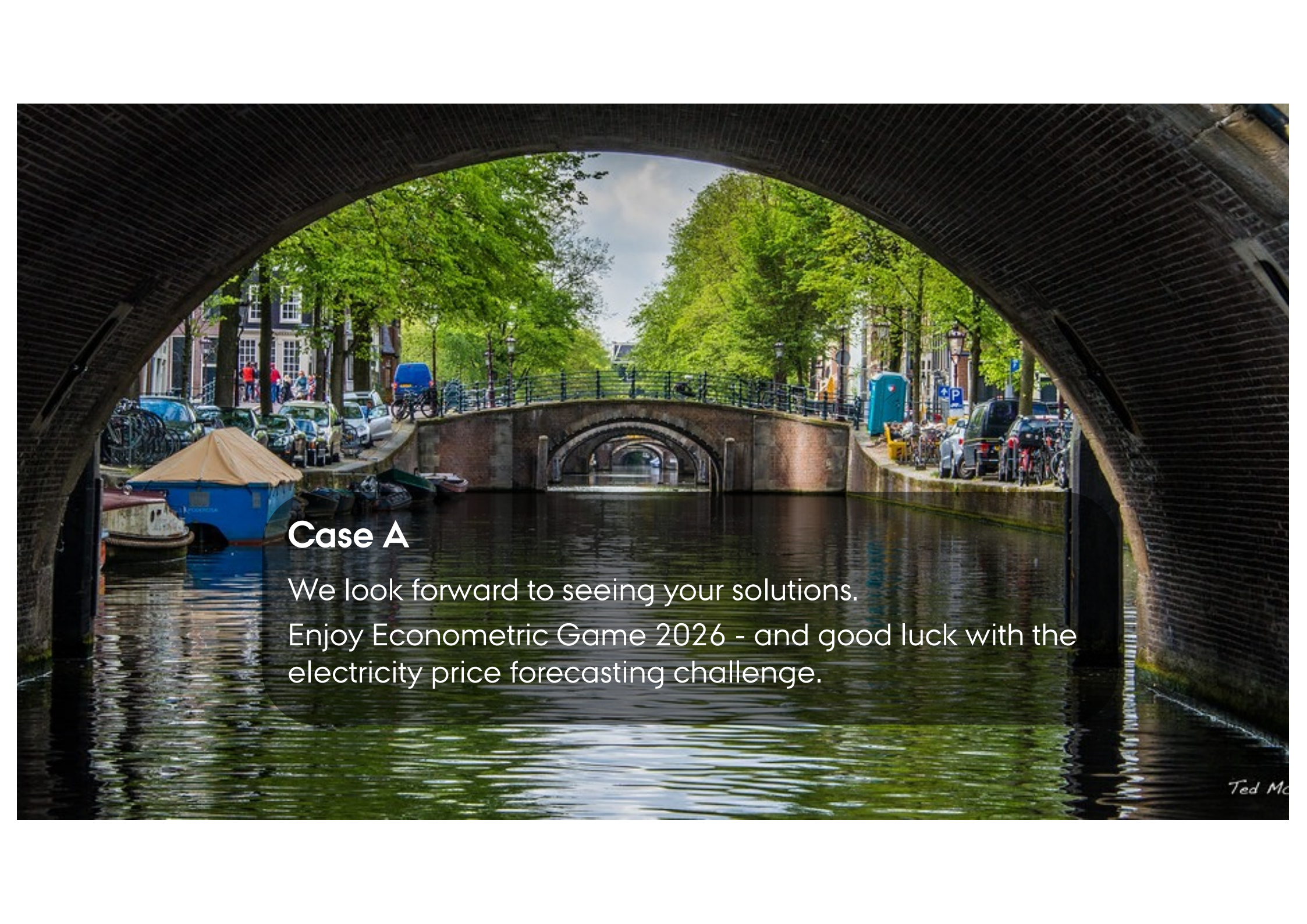
Not allowed

- Cloud computing resources
- Writing the report's explanations
- Writing analysis or conclusions
- Replacing the teams own academic judgements

Report should include a **list of all the software** used for the solving the case.

Report should provide a **statement** exactly how ChatGPT or similar tools was used during the competition.

Study carefully the rules of the game and the practicalities in the case text.



Case A

We look forward to seeing your solutions.
Enjoy Econometric Game 2026 - and good luck with the
electricity price forecasting challenge.



DEPARTMENT OF ECONOMICS
AND BUSINESS ECONOMICS
AARHUS UNIVERSITY



Center for Research in Energy
ECONOMICS AND MARKETS

ECONOMETRIC GAME 2026

NIELS HALDRUP
PROFESSOR

